UNIT IV

Chapter 12

Basic Medical Instruments and Techniques

Chapter Outline

12.1 Diagnostic and Monitoring Instruments
12.2 Imaging Instruments
12.3 Therapeutic Instruments
12.4 Biomedical Techniques



'Smart contact lens sensor for diabetic and glaucoma diagnosis

Learning Objectives:

- *Understands the principles and working mechanisms of instruments*
- *Learn blood cell counting by haemocytometer*
- *Prepares blood smear and learns the different types of WBC*

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Traditionally doctors employ simple instruments like thermometer, stethoscope, and sphygmomanometer to monitor the health conditions of individuals. Medical technology has blossomed during the last few decades and revolutionized medical practices which are immensely influencing life saving means in emergency care of patients. Modern hospitals are established with sophisticated medical instruments like autoanalysers, electrocardiograms, electroencephalograms, ultrasound scanners, CT scanners, NMRI, Isotope imaging and a variety of modern complex equipments for diagnosis and treatment of patients. In medical therapeutics both diagnosis and treatment are two important aspects. Diagnosis involves identifying

- **Haematology**, is the branch of medicine concerned with the study of the cause, prognosis, treatment, and prevention of diseases related to blood.
- **Toxicology** is a discipline that involves the study of the adverse effects of chemical substances on living organisms and the practice of diagnosing and treating exposures to toxins and toxicants.

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